Unsupervised Bayesian Subpixel Mapping Autoencoder Network for Hyperspectral Images

Yuan Fang, Yuxian Wang, Linlin Xu, *Member, IEEE*, Yujia Chen, Alexander Wong, *Senior Member, IEEE*, and David A. Clausi, *Senior Member, IEEE*

***Abstract*— Unsupervised subpixel mapping (SPM) of hyperspectral image (HSI) is a challenging task due to the difficulties to integrate different prior information and model constraints into a coherent framework. This article presents a Bayesian neural network for unsupervised HSI SPM, which has the following characteristics. First, the deep image prior (DIP) achieved by a fully convolutional neural network (FCNN) is used to model the spatial correlation efficiently and adaptively in the subpixel label domain. Second, a discrete spectral mixture model (DSMM) is designed to leverage the forward model for enhanced SPM. Third, an autoencoder architecture is designed to integrate the FCNN and the DSMM to allow efficient unsupervised representational learning using both data and knowledge. Fourth, an expectation-maximization approach is designed to solve the resulting maximum a posteriori (MAP) problem, where a purified means approach extracts endmembers, and the gradient descent approach updates FCNN parameters for subpixel label estimation. Comparative experiments on both real and simulated HSIs demonstrate that the proposed method outperforms other state-of-the-art methods in terms of both numerical accuracies and visual SPM results.**

***Index Terms*— Bayesian framework, deep image prior (DIP), fully convolutional neural network (FCNN), hyperspectral image (HSI), spatial correlation, subpixel mapping (SPM) unmixing.**

I. Introduction

HYPERSPECTRAL imaging is a rapidly growing remote-sensing technique and has been widely used for applications, such as ground target classification [1], [2], [3], agricultural management [4], and environmental monitoring [5], [6]. However, due to the tradeoff between spectral resolution and spatial resolution in hyperspectral images (HSIs), pixels in HSIs usually contain spectral contributions from multiple materials. Subpixel mapping (SPM) creates a label map with a finer spatial resolution by dividing the original mixed pixel into several subpixels [7], which benefits a wide range of applications, such as environmental monitoring [8], [9], [10], target detection [11], and rural land cover objects [12].

Efficient unsupervised SPM approaches are in demand since ground-truth data of HSIs is limited. However, it is very challenging to achieve the unsupervised SPM of HSIs because it is an ill-posed problem where prior information, such as spatial correlation effect and spectral mixture effect, is critical for establishing effective model constraints [13], [14], [15]. Nevertheless, efficient usage of spatial correlation modeling and forward HSI generation model is insufficiently studied in current deep learning (DL)-based SPM methods. Most current DL-based SPM methods, for example, [16], [17], [18], are supervised approaches where the ground-truth data regarding subpixel label information are required to train SPM algorithms. Due to the supervised nature of these algorithms, they do not rely much on the knowledge of the spectral mixture model (SMM) to estimate the subpixel labels. However, given the difficulty to obtain trustworthy ground-truth information, it is critical to study the unsupervised SPM approaches that can adaptively estimate subpixel labels using spatial information and the knowledge in the SMM in a unified framework. To achieve this, three key issues are identified to be addressed.

First, although efficient modeling of the spatial prior is essential for accurate SPM, it is not sufficiently researched. Spatial dependence assumption (SDA)-based priors assume that close pixels have a higher correlation than distant ones [19], such as the spatial attraction SPM (SASM) [7], [20], [21], pixel swapping algorithm (PSSM) [22], and genetic algorithm SPM [23], [24]. However, SDA prior is limited by the sampling scale on HSI pixels [25], and near pixels are not always more correlated than distant ones. Markov random field (MRF) prior is more precise than SDA priors [26], [27], but the fixed priors cannot sufficiently characterize an HSI with the geographically realistic spatial distribution [28]. DL-based methods make use of neural networks to achieve a learnable prior from training samples. Both fully-connected networks [11], [29] and convolutional neural networks (CNNs) [30] have been adopted to solve the SPM problem. However, heavily relying on the fixed kernels, these methods are insufficient in adaptively characterizing the realistic spatial correlation in subpixel fields [19]. The structure of fully CNNs (FCNNs) is capable to capture image statistical spatial information and to impose an adaptive deep image prior (DIP) to restore high-quality images from low-quality

Manuscript received 25 August 2022; revised 17 January 2023 and 29 March 2023; accepted 28 April 2023. Date of publication 11 May 2023; date of current version 23 May 2023. This work was supported in part by the Natural Sciences and Engineering Research Council of Canada (NSERC) under Grant RGPIN-2017-04869, Grant DGDND-2017-00078, Grant RGPAS2017-50794, and Grant RGPIN-2019-06744. *(Yuan Fang and Yuxian Wang contributed equally to this work.) (Corresponding author: Linlin Xu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3272270

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images [31]. Recent publications show the effectiveness of DIP for image restoration [32], [33], [34]. However, FCNN-based approaches leveraging the DIP have not been adapted to HSIs for enhanced SPM. Therefore, how to integrate the DIP into SPM is an important research issue.

Second, although it is critical to integrate the spectral information of HSIs into a deep neural network for enhanced SPM, such a spectral constraint has not been efficiently leveraged for HSI SPM. The challenges are two folds. First, it is challenging to make a deep neural network learn from the spectral model in a coherent way. Second, the most commonly used SMM cannot build a direct connection of the subpixel labels with abundance, endmembers, and the HSI, which makes SPM difficult. So, most DL-based SPM methods ignore the spectral models and heavily rely on ground-truth labels. However, the failure of using the spectral model to constrain SPM leads to heavy supervision for accurate estimation. Therefore, it is critical to design a forward SMM that can not only effectively describe the relationship among subpixel labels, abundances, endmembers, and the HSI, but can also be integrated into a deep neural network. We use a discrete SMM (DSMM) instead of the traditional SMM to build the relationship map from the subpixel labels and endmembers to the HSI, which allows endmember constraints to be imposed to enhance the SPM. To integrate the forward DSMM with the deep neural network, we adopt an encoder–decoder architecture that accommodates the designed DSMM as the decoder and an FCNN as the encoder. Such strategy leveraging both the deep neural network and forward model has succeeded in several image restoration tasks [35], as well as in our previous work on the HSI unmixing [36]. Similarly, the unrolling or unfolding networks [15] integrate model- and learning-based methods by combining the known forward model and deep networks and training the end-to-end network for HSI fusion [13], [14] and super-resolution [15]. Nevertheless, to the best of our knowledge, it has not been studied in the SPM scenario which is more challenging.

Third, following the above discussion, it is promising to integrate the spatial prior (i.e., DIP) and the forward model (i.e., DSMM) in a coherent framework for enhanced unsupervised SPM. However, for the SPM task, from the optimization perspective, it is still very difficult to estimate all unknown variables (i.e., the subpixel labels, abundance, and endmembers) simultaneously, which cannot be achieved by simply training the network like the unfolding networks [13], [14], [15]. The Bayesian method is used widely to solve the image inverse problem by integrating prior information with the posterior distributions given the image observation [37], [38]. For example, the MRF has been incorporated as the prior for HSIs in the Bayesian framework [3]. The SPM problem can be driven as a maximum a posteriori (MAP) problem in a Bayesian framework and solved with the expectation–maximization (EM) approach. Therefore, it is important to design an EM algorithm tailored to the SPM problem in the Bayesian framework. In this article, a Bayesian SPM autoencoder network (BSMAN) for unsupervised SPM of HSIs is proposed, where the physical forward model (i.e., the SMM) is integrated with a learnable FCNN in a Bayesian framework. The BSMAN with an encoder–decoder architecture has the following characteristics.

1) The DIP achieved by a well-designed skip-connected FCNN is used to model the spatial correlation efficiently and adaptively in the subpixel label domain. The DIP maps the low-resolution HSI to class-memberships of subpixels, and servers as the encoder component in the overall architecture of the proposed method.
2) A DSMM is designed to leverage the forward model for enhanced SPM. This DSMM model first maps the subpixel class membership to coarse-pixel abundance and then reconstructs the input HSI using the coarse-pixel abundance and estimated endmembers. This DSMM model serves as part of the decoder component in the overall architecture of the proposed method.
3) An autoencoder architecture is designed to integrate the DIP model with the DSMM to allow efficient unsupervised representational learning using both data and knowledge in the SMM.
4) An EM approach is designed to solve the resulting MAP problem, where a purified means approach is designed to extract endmembers, and the gradient descent approach is used to update FCNN parameters for subpixel label estimation.

Comparative experiments on both simulated and real HSI demonstrate that the proposed approach can generate subpixel labels with higher accuracy than other traditional and state-of-the-art approaches by exploring the spatial correlation effect in the HSI. The remainder of the article is organized as follows. Section III formulates the SPM problem into a MAP problem in a Bayesian framework. The design of the network is detailed in Section IV. Section V introduces the optimization scheme of the proposed BSMAN. Section VI shows experimental results on both simulated and real HSIs.

II. Related Study

A. Deep Image Priors

DIP is the prior information built in the randomly initialized FCNN instead of other learned priors for solving inverse problems of images [31]. Recent publications show the effectiveness of DIP for image restoration [32], [33], [34], for example, HSI unmixing [36], super-resolution [39], image inpainting [35], and denoising [40], [41]. In DIP-based image restoration algorithms, a learnable deep neural network is adopted as an encoder to learn a high-quality image generator from random noise, and a task-specific forward model for low-quality image reconstruction is integrated as a decoder. The prior information of the high-quality image is captured by the neural network structure and imposed on the optimization framework to solve the inverse problem.

The FCNN-based approach leveraging the DIP has not been adapted to HSIs for enhanced SPM. We use DIP to adaptively model the spatial correlation among subpixels as a learning approach instead of an engineering approach for SPM.

B. Spectral Unmixing

Spectral unmixing (SU) decomposes mixed pixels in an HSI into the spectral signatures of constituent components

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(i.e., endmembers) and their fractional proportions (i.e., abundances) [42]. SU creates abundance maps with the original spatial resolution, while SPM creates a label map with a finer spatial resolution by dividing the original mixed pixel into several subpixels.

For the abundance estimation in SU, FCNN-based algorithm [36] outperforms the traditional approaches that treat abundances pixel independently (such as nonnegative least-squares (NNLS) [43], sparse unmixing by variable splitting and augmented Lagrangian (Sunsal) [44], and fully connected networks [45], [46]). However, FCNN-based approaches leveraging the DIP have not been adapted to HSIs for enhanced SPM.

For the endmember extraction in SU, although the minimum volume simplex constraint (e.g., vertex component analysis (VCA) [47] and PPI [48]) are the most used methods, they can only work under the assumption that the HSI contains pure pixels [42]. "Purified means" approach [49], [50], [51] is achieved by treating the endmember as the mean vector of the "purified" pixels, which provides an effective constraint on endmember extraction and works more effective than traditional methods when pixels are highly mixed.

III. Problem Formulation

We assume that an observed HSI data cube $\mathbf{X}$ has P spectral bands, N pixels containing m rows and n columns, and the term I represents the set of coarse pixel sites in HSI, we denote the observed reflectance of the pixel at site i by $\boldsymbol{x}_i$, which is a $P \times 1$ vector. Then the HSI can be expressed as $\mathbf{X} = \{\boldsymbol{x}_i | i = 1, 2, \ldots, m \times n\}$. The term J represents the set of subpixel positions within each coarse pixel, which contains a total of c^2 positions. For example, when $c = 2$, a coarse pixel is divided into 2×2 subpixels, and the coarse HSI containing $m \times n$ pixels corresponds to a fine HSI with the size of $2m \times 2n$ subpixels.

Assuming that the HSI covers K classes, the SPM aims to infer subpixel labels $\mathbf{L} = \{\boldsymbol{l}_{i,j} | i \in I, j \in J\}$ in the HSI, where $\boldsymbol{l}_{i,j}$ is a one-hot $K \times 1$ vector where the nonzero element defines the class of the subpixel. The discrete abundance $\mathbf{S} = \{\boldsymbol{s}_i | i = 1, 2, \ldots, m \times n\}$ of coarse pixels is determined by the proportions of subpixels labels, which is expressed as follows:

$$\boldsymbol{s}_i = \frac{\sum_{j=1}^{c^2} \delta\big(\max(\boldsymbol{l}_{i,j}), k\big)}{c^2} \tag{1}$$

where $\max(\boldsymbol{l}_{i,j})$ returns the index of the nonzero element (i.e., the class label), and $\delta(u, v)$ is the Kronecker delta function, where $\delta(u, v) = 1$ for $u = v$ and $\delta(u, v) = 0$ otherwise. For example, as illustrated in Fig. 1, when $c = 2$, there is one "tree" subpixel and three "flower" subpixels. Then the discrete abundance $\boldsymbol{s}_i$ corresponding to the coarse pixel is a 2×1 vector written as $[1/4; 3/4]$.

Then, a coarse pixel $\boldsymbol{x}_i$ can be formulated as a linear combination of K endmembers $\mathbf{A} = \{\boldsymbol{a}_k | k = 1, 2, \ldots, K\}$ weighted by discrete abundances plus noise $\boldsymbol{n}_i$, which is a DSMM

$$\boldsymbol{x}_i = \sum_{k=1}^{K} \boldsymbol{a}_k \frac{\sum_{j=1}^{c^2} \delta\big(\max(\boldsymbol{l}_{i,j}), k\big)}{c^2} + \boldsymbol{n}_i. \tag{2}$$

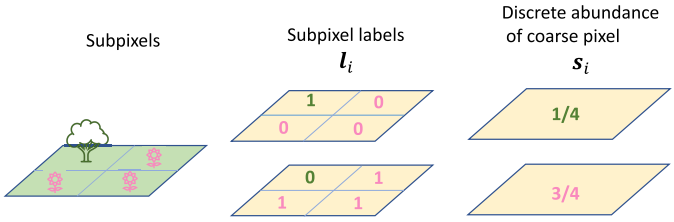


Fig. 1. Illustration of the relationship between the subpixel labels and discrete abundances in the coarse pixel.

The noise distribution is assumed to satisfy a Gaussian model as follows:

$$p(\boldsymbol{n}_i) = \frac{1}{\sqrt{(2\pi)^P}} \exp\left(-\frac{1}{2}\boldsymbol{n}_i{}^T \boldsymbol{n}_i\right). \tag{3}$$

SPM in a Bayesian framework can be achieved by maximizing the posterior distribution of $\mathbf{L}$ given $\mathbf{X}$, that is,

$$p(\mathbf{L} \mid \mathbf{X}) \propto p(\mathbf{X} \mid \mathbf{L}) p(\mathbf{L}). \tag{4}$$

Based on the above formulation, we solve the ill-posed SPM problem in a Bayesian framework. The main characteristics of the proposed method are summarized below.

1) The spatial prior of subpixel labels $p(\mathbf{L})$ is modeled by DIP that is achieved by a designed skip-connected FCNN, as detailed in Section IV-A.
2) The data likelihood $p(\mathbf{X} \mid \mathbf{L})$ is modeled by a Gaussian distribution, where the designed DSMM in (2) models the mean of $\mathbf{X}$, as detailed in Section IV-B.
3) An efficient EM algorithm-based optimization scheme is designed and implemented in Section V-B for solving a MAP problem.
4) The gradient descent approach is used to update FCNN parameters, as detailed in Section V-C.
5) Endmembers $\mathbf{A}$ are extracted with a purified means approach [49], as detailed in Section V-D.

IV. Bayesian SPM Autoencoder Network

The proposed BSMAN has an encoder–decoder structure. The implementation of BSMAN is illustrated in Fig. 2. The encoder is a skip-connection FCNN designed for the estimation of subpixel labels $\{\boldsymbol{l}_{i,j}\}$, where DIP is used to model the spatial correlation of the subpixel label field. The encoder maps the input (i.e., coarse HSI $\{\boldsymbol{x}_i\}$) to the subpixel labels $\{\boldsymbol{l}_{i,j}\}$. The decoder has two parts. One part is the forward model which maps the subpixel labels $\{\boldsymbol{l}_{i,j}\}$ to the discrete abundances. The other part of the encoder reconstructs the HSI with class proportions and endmembers extracted from the coarse HSI.

A. Prior of Subpixel Labels

The prior of $\{\boldsymbol{l}_{i,j}\}$ is expressed as a Gaussian distribution as follows [35], [36]:

$$p(\boldsymbol{l}_{i,j}) = \frac{1}{w} \exp\left(-\left\|\boldsymbol{l}_{i,j} - E(\boldsymbol{l}_{i,j})\right\|^2\right) \tag{5}$$

where $E(\boldsymbol{l}_{i,j})$ is the expectation of $\boldsymbol{l}_{i,j}$. In this article, $E(\boldsymbol{l}_{i,j})$ is achieved by the encoder component (i.e., an FCNN) of

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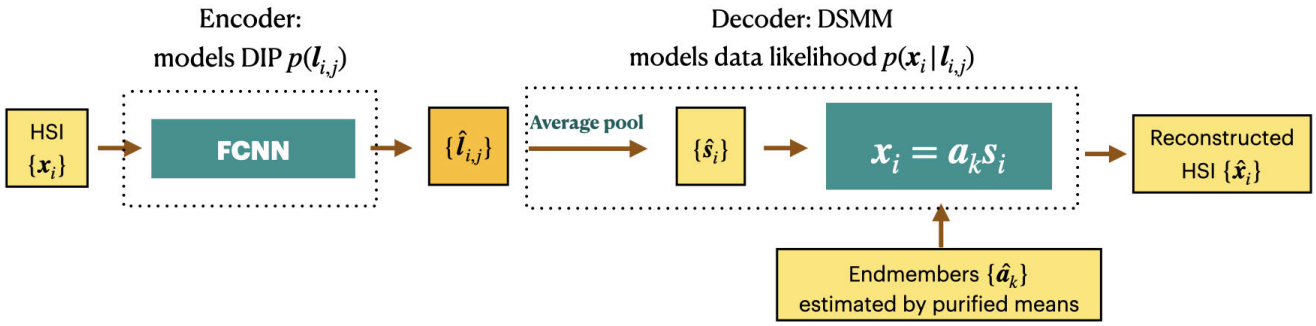


Fig. 2. SPM framework. The encoder is a skip-connection FCNN designed for the estimation of soft labels of subpixels $\{\boldsymbol{l}_{i,j}\}$, where DIP is used to model the spatial correlation of the label field. The decoder has two parts. One part is the forward downsampling model which maps soft labels of subpixels to the class proportions $\boldsymbol{s}_i$. The other part of the encoder reconstructs the HSI $\hat{\mathbf{X}}$ with $\boldsymbol{s}_i$ and endmembers $\mathbf{A}$ extracted from the HSI $\mathbf{X}$.

the proposed framework shown in Fig. 2. The FCNN exploits the DIP [35] to model the spatial correlation in the subpixel label domain. This prior encourages pixels that are spatially correlated with the same class. Using $f(\cdot)$ to represent the FCNN forward propagation, the expected $\boldsymbol{l}_{i,j}$ is written as

$$E(\boldsymbol{l}_{i,j}) = f(\boldsymbol{z}_i, \boldsymbol{\beta}) \tag{6}$$

where $\boldsymbol{z}_i$ is the input random noise and $\boldsymbol{\beta}$ is the set of network parameters including weights and biases. The nonnegative and the sum-to-one constraint is achieved with a "softmax" activation. In (5), $\boldsymbol{l}_{i,j}$, as the output of the last softmax layer, refers to the soft labels of subpixels instead of the hard labels.

B. Data Likelihood

Based on (2) and (3), the data likelihood is formulated as

$$p\left(\boldsymbol{x}_i \mid \boldsymbol{l}_{i,j}\right) = \frac{1}{z}\exp\left\{-\frac{1}{2}\left\|\boldsymbol{x}_i - \left(\sum_{k=1}^{K}\boldsymbol{a}_k \frac{\sum_{t=1}^{c^2}\delta\left(\max(\boldsymbol{l}_{i,t}), k\right)}{c^2}\right)\right\|^2\right\} \tag{7}$$

where the exponential part is the reconstruction error between the original HSI pixel $\boldsymbol{x}_i$ and the reconstructed one, which is implemented by the decoder part in the proposed framework in Fig. 2.

V. Model Optimization

A. MAP Estimation

The SPM problem in (4) can be solved by the MAP approach by maximizing the posterior distribution of $\mathbf{L}$ given the observed HSI $\mathbf{X}$ and the model parameters (i.e., endmembers) $\mathbf{A}$ as follows:

$$\hat{\mathbf{L}} = \arg\max_{\mathbf{L}} \{p(\mathbf{L} \mid \mathbf{X}, \mathbf{A})\}. \tag{8}$$

Maximizing $p(\mathbf{L} \mid \mathbf{X}, \mathbf{A})$ is equivalent to minimizing its negative logarithm likelihood. Then, the objective function can be written as

$$\min_{\mathbf{L}} \sum_{i=1}^{N}\sum_{j=1}^{c^2}\left\{\left\|\boldsymbol{l}_{i,j} - E\left(\boldsymbol{l}_{i,j}\right)\right\|^2\right\} + \alpha\sum_{i=1}^{N}\left\{\left\|\boldsymbol{x}_i - \left(\sum_{k=1}^{K}\boldsymbol{a}_k \frac{\sum_{j=1}^{c^2}\delta\left(\max(\boldsymbol{l}_{i,j}), k\right)}{c^2}\right)\right\|^2\right\} \tag{9}$$

where α is a weighting parameter. Both endmembers $\{\boldsymbol{a}_k\}$ and subpixel labels $\{\boldsymbol{l}_{i,j}\}$ are unknown variables, making the SPM problem ill-posed. In the EM algorithm, $\boldsymbol{l}_{i,j}$ is estimated by $E(\boldsymbol{l}_{i,j})$. Replacing $\boldsymbol{l}_{i,j}$ with $E(\boldsymbol{l}_{i,j})$, the objective function can be reformulated as follows [35]:

$$\min_{\mathbf{L}} \sum_{i=1}^{N}\left\{\left\|\boldsymbol{x}_i - \left(\sum_{k=1}^{K}\boldsymbol{a}_k \frac{\sum_{j=1}^{c^2}\delta\left(\max(E(\boldsymbol{l}_{i,j})), k\right)}{c^2}\right)\right\|^2\right\}. \tag{10}$$

This objective function has the following characteristics.

1) The EM algorithm estimates parameters by treating $\{\boldsymbol{l}_{i,j}\}$ as missing observations and $\{\boldsymbol{a}_k\}$ as model parameters and iteratively updates the estimation of $\{\boldsymbol{l}_{i,j}\}$ and $\{\boldsymbol{a}_k\}$, as illustrated in Section V-B.
2) We use $E(\boldsymbol{l}_{i,j})$ as the estimation of $\boldsymbol{l}_{i,j}$, where $E(\boldsymbol{l}_{i,j})$ is modeled by an FCNN. $\{\boldsymbol{x}_i\}$ is used as the input to the FCNN. Once the FCNN is trained, we obtain $\hat{\boldsymbol{l}}_{i,j} = f(\boldsymbol{l}_{i,j})$, as illustrated in Section V-C.
3) When estimating parameters in the FCNN for obtaining $E(\boldsymbol{l}_{i,j})$, we use a reconstruction loss based on $||\boldsymbol{x}_i - \hat{\boldsymbol{x}}_i||^2$, which incorporates a forward model to constrain meaningful $\boldsymbol{l}_{i,j}$ estimation, as illustrated in Section V-C.

B. EM Iteration

The EM algorithm optimizes the incomplete data problem by iterating between the E-step (i.e., estimation of model parameters given missing observations) and M-step (i.e., estimation of missing observations given the model parameters) [52]. The main steps in EM algorithm to estimate $\{\boldsymbol{l}_{i,j}\}$ and $\{\boldsymbol{a}_k\}$ are summarized as follows.

1) *Initialization:* Set the initial value for $\{\boldsymbol{a}_k\}$. The endmember of each class can be obtained either by manually selecting from the coarse HSI or by other endmember extraction approaches such as VCA [47].
2) *E-Step:* Given endmembers $\{\boldsymbol{a}_k\}$, estimate subpixel soft labels $\{\boldsymbol{l}_{i,j}\}$ by optimizing an FCNN, as introduced in Section V-C.

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3) *M-Step:* Given subpixel soft labels $\{\boldsymbol{l}_{i,j}\}$, update endmembers $\{\boldsymbol{a}_k\}$. Endmembers $\{\boldsymbol{a}_k\}$ are estimated with the purified means approach [49], which is described in Section V-D.

C. Subpixel Labels Estimation by FCNN Training

The objective of E-step of the EM algorithm introduced in Section V-B is to obtain the estimated subpixel labels $\hat{\mathbf{L}}$ which is achieved by an FCNN in (6). The FCNN structure is shown in Fig. 3.

The parameters of the FCNN, that is, $\boldsymbol{\beta}$, need to be optimized and fixed to estimate $E(\boldsymbol{l}_{i,j})$. Here, we construct the following objective function to estimate $\boldsymbol{\beta}$:

$$\min_{\boldsymbol{\beta}} \sum_{i=1}^{N} \left\{ \left\| \boldsymbol{x}_i - \left(\sum_{k=1}^{K} \boldsymbol{a}_k \frac{\sum_{j=1}^{c^2} \delta(\max(f(\boldsymbol{x}_i, \boldsymbol{\beta})), k)}{c^2} \right) \right\|^2 \right\}. \tag{11}$$

The Adam stochastic optimizer [53] is adopted to estimate $\boldsymbol{\beta}$.

D. Endmembers Update by Purified Means

The endmember update in the M-step of the iterative EM algorithm introduced in Section V-B is achieved by the purified means approach, where the purified pixel $\boldsymbol{y}_i^k$ is represented as [49]

$$\boldsymbol{y}_i^k = \boldsymbol{x}_i^k - \sum_{q \neq k}^{K} \frac{\sum_{j=1}^{c^2} \delta\left(\max(\boldsymbol{l}_{i,j}), q\right)}{c^2} \boldsymbol{a}_q. \tag{12}$$

By solving the following objective function:

$$\{\hat{\boldsymbol{a}}_k\} = \arg\min_{\boldsymbol{a}_k} \sum_{k=1}^{K} \left\| \boldsymbol{a}_k - \frac{1}{N} \sum_{i}^{N} \boldsymbol{y}_i^k \right\|^2 \tag{13}$$

the endmember matrix $\{\hat{\boldsymbol{a}}_k\}$ can be iteratively updated using their conditional expectations.

E. Summary of Complete Algorithm

The input of the whole BSMAN algorithm is the coarse HSI as well as two hyperparameters (i.e., number of classes and iteration numbers). The output of BSMAN is the SPM label map. The entire unsupervised process is achieved with EM iterations without requiring any training datasets. Based on the EM steps described in Section V-B, the complete algorithm used for solving BSMAN can be achieved which is summarized in Algorithm 1.

Algorithm 1 BSMAN

Input: HSI $\mathbf{X}$, numbers of classes K, and iteration numbers τ
Output: Class labels of subpixels $\{\boldsymbol{l}_{i,j}\}$ **Initialization**: $t := 1$, $\{\boldsymbol{a}_k^{(0)}\}$

```
While t ⩽ τ do
  E-step:
    estimate β in FCNN,
    estimate {l_{i,j}} ← fcnn({x_i}, {a_k}) in Eq. (6)
  M-step:
    for k = 1, 2, ..., K
      estimate {y_i^k} according to Eq. (12)
      estimate a_k using {y_i^k} according to Eq. (13)
    end for
end while
```

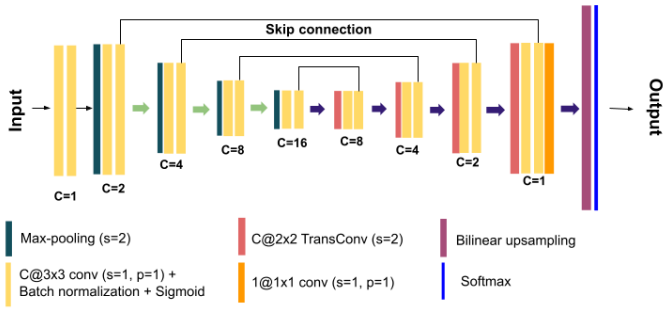


Fig. 3. FCNN structure in the BSMAN framework. C represents the number of convolution channels.

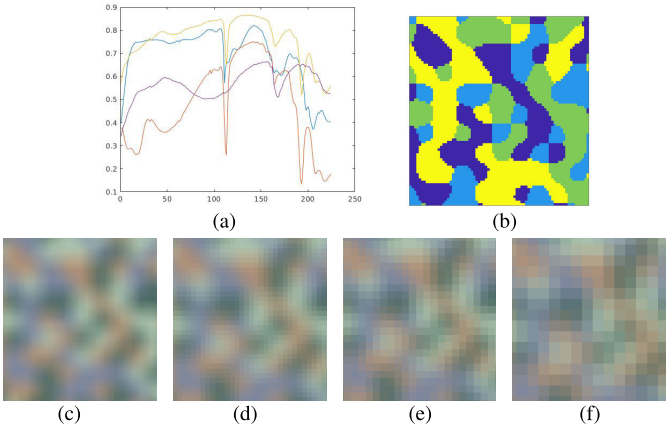


Fig. 4. False color images of simulated HSIs. (a) Endmember. (b) Label. (c) Simu, fine. (d) Simu1, $c = 2$. (e) Simu2, $c = 3$. (f) Simu3, $c = 4$.

VI. Experiments

A. BSMAN Implementation

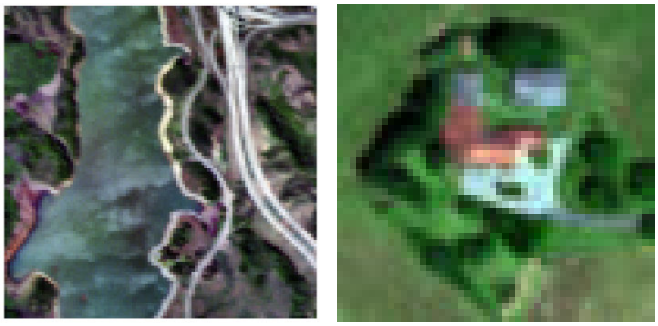
To implement BSMAN, we use the coarse HSI as the input of the FCNN, and the output of the FCNN is the soft subpixel labels. The forward model is implemented with the average pooling to map soft subpixel labels to a discrete abundance of coarse pixels, and a linear combination to reconstruct the HSI using discrete abundances and endmembers. The FCNN is implemented with a U-Net-type "hourglass" architecture with skip connection [35] to model mapping from the input to soft labels of subpixels.

B. Datasets

Six images are used to evaluate the proposed BSMAN, including three simulated HSIs, two real HSIs, and one multispectral image dataset.

1) Simulated HSIs: In this experiment, the 78 × 78 sized fine HSIs are simulated with four endmembers with 200 bands. Each pixel in the simulated HSI is a mixture of four endmembers shown in Fig. 4. Coarse pixels are created using the four endmembers multiplied by four abundance maps following LSMM. Abundance maps are generated by first being divided into 8 × 8 sized homogeneous blocks of one of the four endmembers and then degrading the blocks by applying a spatial low-pass filter with the size of 9 × 9 pixels [36].

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(a) (b)

Fig. 5. RGB images of real HSIs with fine resolutions (note: the input to the network are the HSIs with coarse resolutions, which are obtained by applying the mean filter to the fine-resolution HSIs). (a) Jasper. (b) Saint.

The resulting I is further degraded by zero-mean Gaussian noise (SNR = 40 dB). The label of each pixel is determined by the dominant endmember in the coarse pixel. The coarse HSIs at three degradation scales (i.e., 2, 3, and 4) are obtained by applying mean spatial filters with the kernel sizes of 2×2, 3×3, and 4×4 pixels separately to downsample the fine HSI. As a result, three HSIs with the sizes of 39×39, 26×26, and 19×19 are simulated.

2) Jasper Ridge HSI Dataset: Jasper Ridge is an HSI dataset with 100×100 pixels at 198 spectral bands from 0.38 to 2.5 μm [36]. It contains four endmembers, that is, "road," "soil," "water," and "tree." The ground truth for the dataset is generated by Zhu [54]. The label of each subpixel is determined by its dominant endmember. The red, green and blue (RGB) color composition of the fine HSI is shown in Fig. 5(a). The input coarse HSI with the size of 50×50 is obtained by applying a 2×2 mean filter to the original HSI.

3) Saint Andre HSI Dataset: The Saint Andre HSI contains 50×50 pixels, composed of 415 spectral bands ranging from 0.40 to 2.40 μm [55]. Six endmembers (i.e., "tree," "grass," "soil," "road," "building 1," and "building 2,") were extracted based on prior knowledge of the scene. The label of each subpixel is determined by its dominant endmember. The RGB color composition of the HSI is shown in Fig. 5(b). The input coarse HSI with the size of 25×25 is obtained by applying a 2×2 mean filter to the original HSI.

4) Landsat Imagery Dataset: An HSI-like image was obtained by combining 11 Landsat-8 images over Gulin, Sichuan, China, with cloud cover lower than 10%. We selected a subarea from the whole scene. By removing the panchromatic bands and two damaged bands, the size of the subarea dataset is $40 \times 40 \times 108$ with 30-m spatial resolution. The dataset was used in the previous study on HSI classification [3]. There are five land cover types, that is, road, residential area, forestland, terrace, and farmland. A 1-m-resolution UAV image of the same area is shown in Fig. 6(b). We aim to obtain a label map with 7.5-m resolution given the 30-m Landsat image with 4× SPM scale. The 1-m-resolution UAV image is used as the reference.

C. Experimental Setup

1) Methods Compared: The proposed method is compared with several traditional SPM methods including GAAI [24], SPMLM [56], SASM [7], PSSM [57], RBF [21], and SPMSS [58]. For these compared methods, RBF used the fraction images as input, and other methods used HSIs as input initially. Since we compare all methods in an unsupervised scenario, instead of using the true or manually selected endmembers and abundances, we used the VCA algorithm [47] to extract initial endmembers and the fast NNLS method [59] to obtain initial abundances in the unmixing part of all these compared methods.

2) Numerical Measures: Three kinds of accuracy indices are used to evaluate the quantitative performance of SPM algorithms on real HSIs: the accuracy of each class, the overall

TABLE I
HYPERPARAMETERS SETTING

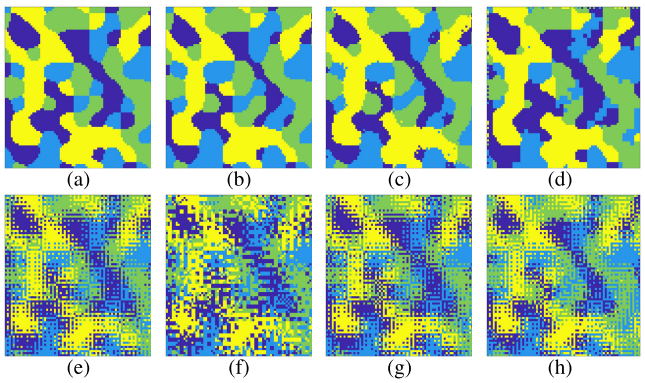
Dataset	input HSI Size	Scale	LR	Epochs	EM	Skips
Simu 1	$39 \times 39 \times 200$	2	0.0001	20	70	4
Simu 2	$26 \times 26 \times 200$	3	0.0001	20	100	4
Simu 3	$19 \times 19 \times 200$	4	0.0001	10	100	4
Jasper	$50 \times 50 \times 224$	2	0.0002	200	50	4
Saint	$25 \times 25 \times 415$	2	0.001	300	20	3
Landsat	$40 \times 40 \times 108$	4	0.0001	1000	10	4

*Simu 1, 2 and 3 refer to simulated HSIs by degrading the original HSI with $c = 2, 3, 4$.
*The fourth column is the learning rate (LR) for FCNN training in E-step.
*The fifth column is the number of epochs for FCNN training in E-step.
*The sixth column is the number of EM iteration.
*The seventh column is the number of the skip-connections in the FCNN, determined by the network depth.



(a) (b)

Fig. 6. True color images of the Landsat scene. (a) Low-resolution (30 m) true color Landsat-8 imagery. (b) High-resolution (0.2 m) UAV imagery that helps the acquisition of the ground-truth map.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. SPM results (subpixel label maps and their OA) of different methods on the simulated dataset with $c = 2$. (a) Label. (b) BSMAN. (c) GAAI. (d) SPMLM. (e) SASM. (f) PSSM. (g) RBF. (h) SPSMM.

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TABLE II
SIMULATED HSI DATA: KAPPA COEFFICIENT AND OA

Methods	c = 2		c = 3		c = 4	
	Kappa	OA (%)	Kappa	OA (%)	Kappa	OA (%)
SPMSS	0.5294	64.73	0.5868	69.05	0.6286	72.21
RBF	0.5526	66.50	0.5868	69.04	0.6505	73.81
PSSM	0.4778	60.90	0.5056	62.92	0.4198	56.22
SASM	0.5529	66.52	0.5920	69.40	0.4936	61.77
SPMLM	0.8169	86.34	0.7795	83.56	0.5674	67.40
GAAI	0.8564	89.27	0.7508	81.36	0.4459	58.14
DIP	**0.8702**	**90.26**	**0.8686**	**90.17**	**0.7077**	**77.98**

TABLE III
JASPER RIDGE HSI DATA: KAPPA COEFFICIENT AND OA

Methods	Kappa	OA (%)
SPMSS	0.3159	50.21
RBF	0.5472	66.22
PSSM	0.5500	67.99
SASM	0.5815	70.24
SPMLM	0.7942	85.59
GAAI	0.7966	85.81
BSMAN	**0.8087**	**86.81**

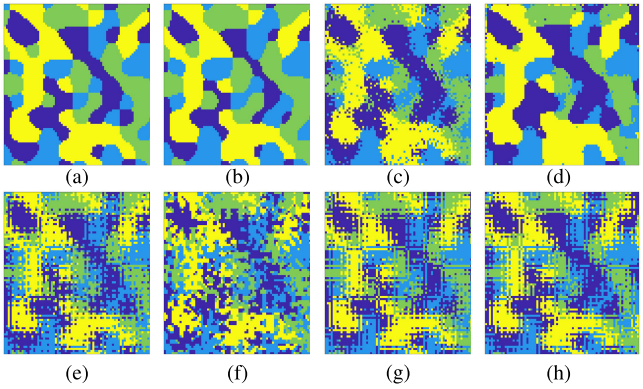


Fig. 8. SPM results (subpixel label maps and their OA) of different methods on the simulated dataset with $c = 3$. (a) Label. (b) BSMAN. (c) GAAI. (d) SPMLM. (e) SASM. (f) PSSM. (g) RBF. (h) SPSMM.

accuracy (OA), and the Kappa coefficient (Kappa). Only the last two indices (i.e., OA and Kappa) are used in the simulated study.

3) Parameter Settings: For the proposed method, hyperparameters including the learning rate, iteration times, and the number of skip connection layers are determined empirically, which are recorded in Table I. Hyperparameters are required to adjust for different HSIs considering HSIs vary in the sensor type, data size, and image complexity. For the parameters of the comparison methods, we mainly refer to the original papers and source codes. About other uncertain parameters, we refer to the selection method of the literature, such as using the L-curve method to determine the value of weighting parameter λ of the SPMLM method and ω of the SPMSS method [56], [58].

D. Simulated Study

Figs. 7–9 show the visual comparison between the proposed method with other methods as three downsampling-scale

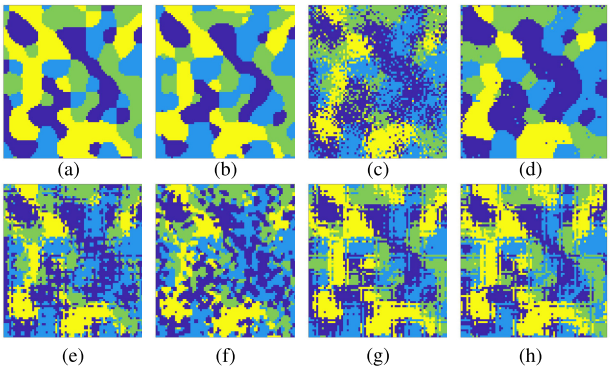


Fig. 9. SPM results (subpixel label maps and their OA) of different methods on the simulated dataset with $c = 4$. (a) Label. (b) BSMAN. (c) GAAI. (d) SPMLM. (e) SASM. (f) PSSM. (g) RBF. (h) SPSMM.

factors. The proposed BSMAN method always generates subpixel label maps with the richest spatial texture details and the least noise. Besides BSMAN, GAAI and SPMLM perform better than the rest of the methods (i.e., SASM, PSSM, RBF, and SPMSS) by delineating the class boundary more clearly and less noisy. SASM, PPSM, and RBF are based on the concept of spatial dependence, assuming that pixels spatially adjacent are more likely to be similar, leading to the blocking effect in the label maps. GAAI shows less blocking effect because it corrects the potential errors in the estimated abundance maps and achieved a global optimization. SPMLM minimizes an objective function using the spectral term to regulate the spatial term by combines utilizes the linear unmixing model and the maximum spatial dependence model. Although SPMSS considers both spatial and spectral correlations, it highly relies on the bilinear and bicubic interpolation results, leading to the grid effect as well. The numerical indices summarized in Table II show consistent results with the visual demonstration. The OA and Kappa of the classification results obtained by the proposed BSMAN are the highest among all tested methods at scale factors two and three. When $c = 4$, although BSMAN gives the most spatial detail information in Fig. 9(b), the performance of BSMAN in terms of OA and Kappa ranks second after SPMLM due to inaccurate label prediction near the edge of the scene. This is caused by the convolutions on small-size images.

E. Test on Real HSIs

1) Test on Jasper Ridge HSIs: Fig. 10 shows the SPM results for the Jasper HSI scene obtained by different SPM methods. The subpixel label map obtained by the BSMAN

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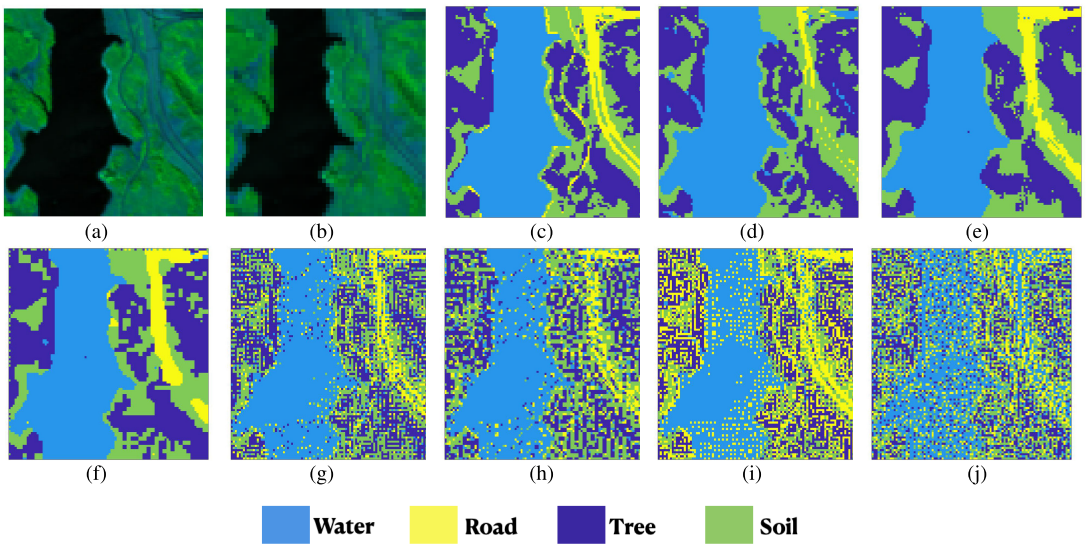


Fig. 10. SPM results (subpixel label maps and their OAs) of different methods on Jasper dataset. (a) Fine image. (b) Input coarse image. (c) Label. (d) BSMAN 86.81%. (e) GAAI 85.81%. (f) SPMLM 85.59%. (g) SASM 70.24%. (h) PSSM 67.99%. (i) RBF 66.22%. (j) SPMSS 50.21%.

TABLE IV
SAINT HSI DATA: KAPPA COEFFICIENT AND OA

Methods	Kappa	OA (%)
SPMSS	0.3013	45.32
RBF	0.2715	51.44
PSSM	0.3541	50.00
SASM	0.3632	50.72
SPMLM	0.4555	64.56
GAAI	0.4317	61.28
BSMAN	**0.6230**	**72.44**

TABLE V
JASPER RIDGE HSI DATA: INDIVIDUAL CLASS ACCURACIES (%)

Methods	Water	Road	Tree	Soil
SPMSS	64.94	41.43	45.78	39.13
RBF	88.45	**71.71**	55.60	48.72
PSSM	91.28	53.25	58.69	54.04
SASM	92.09	59.36	60.38	57.87
SPMLM	99.91	65.74	84.88	73.15
GAAI	99.97	64.28	87.12	71.21
BSMAN	**100.00**	23.24	**90.58**	**83.03**

methods shows the most detailed spatial textural information and achieves the highest numerical accuracy shown in Table IV. Table V shows the individual classification accuracy achieved by different methods on the HSI. The proposed BSMAN performs the best on the class "water," "tree," and "soil." Although the "road" class accuracy is relatively low, the outline of the road is still clearly visible in Fig. 10. We attribute the false negative pixels for road class to the smoothness property of convolutions.

2) Test on Saint Andre HSIs: Fig. 11 shows the SPM results for the Saint HSI scene obtained by different SPM methods. The subpixel label map obtained by the BSMAN method preserves the richest spatial textural information and achieves the highest OA and Kappa coefficient. The Saint scene turns out to be more challenging than the Jasper scene given that all methods in this scene obtain relatively lower SPM accuracy and more noisy label maps. We attribute it to the similar land cover types and higher mixed pixels in the HSI. The class of soil, grass, and tree are commonly mixed in real scenarios, especially for soil and grass at the bottom of this HSI scene. BSMAN successfully identified the dominant class in the HSI and restored the label map illustrated in Fig. 11(b), indicating that BSMAN is more capable of dealing with highly mixed HSI pixels. For individual classes, BSMAN achieves the highest individual class accuracies on classes of soil, road, and building1 displayed in Table VI. However, the accuracy for trees is relatively low because BSMAN treats the shadow part of trees as building1 [see Fig. 11(b)].

3) Test on Landsat Imagery: Fig. 12 shows the SPM results for the Landsat dataset obtained by different SPM methods. As we can see, in Fig. 12(j)–(l), the abundance maps generated by the proposed BSMAN method correlate with the Landsat and UAV images well. In particular, although the building boundaries and curve shape of the road are very vague and poorly defined in the Landsat image [see Fig. 12(i)], the proposed BSMAN method has successfully learned to restore and improve the geometric information by providing better building boundaries in Fig. 12(k) and better curve shape of the road in Fig. 12(j), which matches with the UAV images very well. As a result, the classification map of BSMAN in Fig. 12(g) has a clear curve shape of the road and the accurate boundary of the buildings. Due to limited spatial learning capability, maps of other methods in Fig. 12(a)–(f) show obvious grid effect and tend to mix the road and building together.

F. Discussion

Our proposed method has demonstrated its effectiveness in handling different SPM scales using both simulated HSIs and three real HSI datasets. These real datasets include two downscaled real HSIs, which have the advantage of having no uncertainties resulting from gaps in acquisition time and

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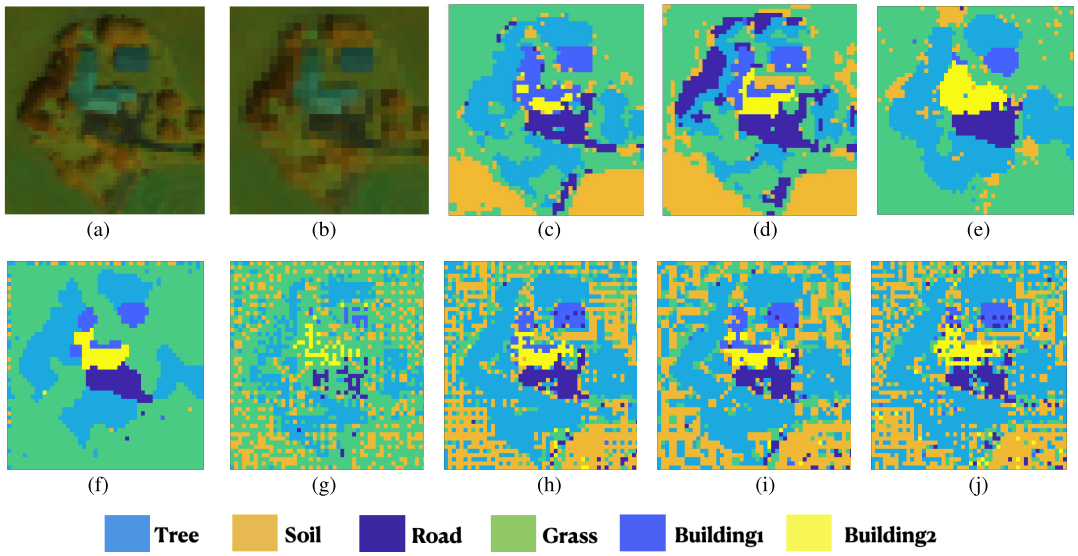


Fig. 11. SPM results (subpixel label maps and their OAs) of different methods on the Saint dataset. (a) Fine image. (b) Input coarse image. (c) Label. (d) BSMAN 72.44%. (e) GAAI 61.28%. (f) SPMLM 64.56%. (g) RBF 51.44%. (h) SASM 50.72%. (i) PSSM 50.00%. (j) SPMSS 45.32%.

TABLE VI

SAINT HSI DATA: INDIVIDUAL CLASS ACCURACIES (%). THE HIGHEST ACCURACY IN EACH ROW IS IN BOLD FORMAT

Methods	Tree	Soil	Road	Grass	Building1	Building2
SPMSS	80.24	0.42	55.28	**86.28**	51.39	83.33
RBF	62.39	36.29	25.47	61.10	0.13	44.44
PSSM	87.52	68.57	59.01	19.77	57.64	63.89
SASM	88.73	70.68	57.76	19.86	55.56	77.78
SPMLM	80.24	0.42	55.28	**86.28**	51.39	83.33
GAAI	**88.91**	1.69	60.25	74.64	35.42	**100.00**
BSMAN	53.55	**84.81**	**84.45**	74.73	**75.00**	77.78

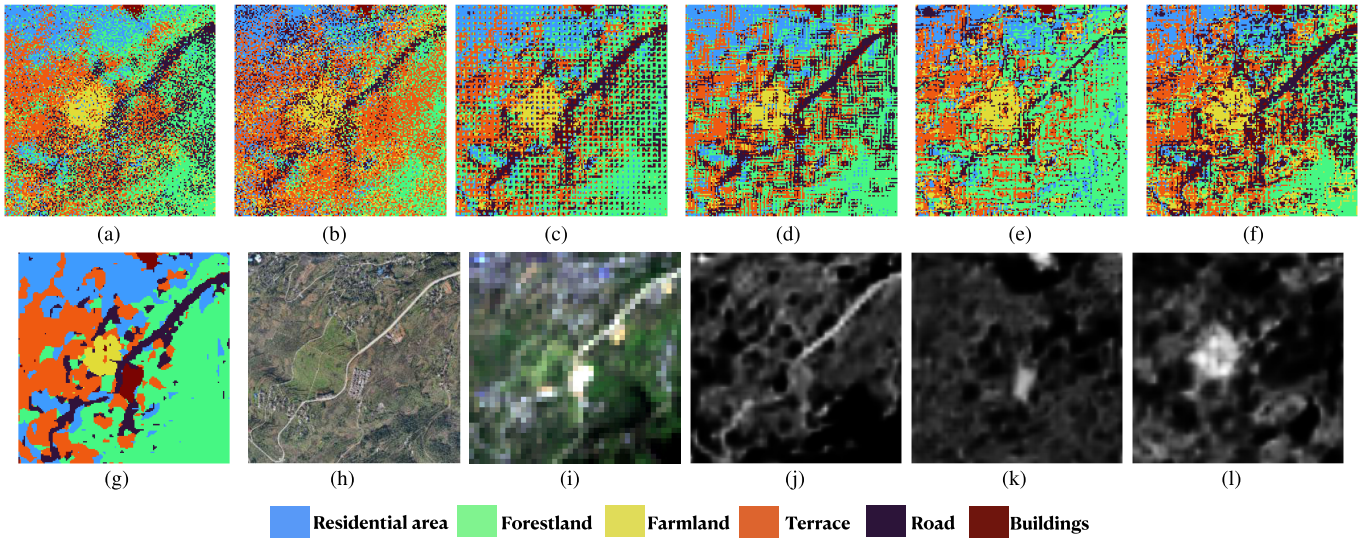


Fig. 12. SPM results of different methods (a)–(g) on the Landsat dataset and the abundance maps (j)–(l) generated by our proposed method. The abundance maps generated by the proposed BSMAN method correlate with (i) Landsat and (h) UAV images well. (i) In particular, although the building boundaries and curve shape of the road are very vague and poorly defined in the Landsat image, (k) proposed BSMAN method has successfully learned to restore and improve the geometric information by providing better building boundaries and (j) better curve shape of the road, which matches with (h) UAV image very well. (g) As a result, the classification map of BSMAN has a clear curve shape of the road and the accurate boundary of the buildings. Due to limited spatial learning capability, maps of other methods (a)–(f) show an obvious grid effect and tend to mix the road and building together.

spatial co-registration, as well as an original Landsat dataset. The latter has the complimentary advantage of having a larger image size and a richer land cover texture, with a higher-resolution UAV image available for the same region.

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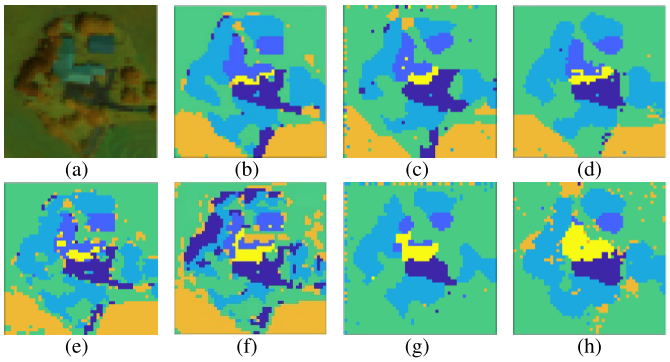


Fig. 13. SPM results (subpixel label maps and their OAs) of different methods with different initialization approaches on the Saint dataset. (a) and (e) Fine image and label map of the Saint dataset, respectively. (b) and (f) Resulting subpixel maps from our proposed BSMAN method. (c) and (g) SPMLM results. (d) and (h) GAAI results. "trueA" displays the result using the true endmember as input, and "VCA" displays the result using the initial endmember from the VCA algorithm. OAs are (b) 86.32%, (c) 80.28%, (d) 82.24%, (f) 72.44%, (g) 64.56%, and (h) 61.28%.

In our proposed BSMAN method, we can initialize the endmembers using both VCA-generated endmembers and true endmembers. However, in most cases, there is no available endmember ground truth data. As an unsupervised approach, it is crucial to achieve effective SPM without having knowledge of true endmembers. Therefore, we are interested in determining how much the performance drops when using VCA-generated endmembers instead of true endmembers, for both our method and others. To this end, we conducted a comparative experiment and present the results in Fig. 13. The figure illustrates that using true endmembers as input greatly enhances model performance for all methods. However, using VCA-generated endmembers significantly harms the SPM accuracy for SPMLM and GAAI, while our BSMAN method can still generate a high-accuracy SPM label map. This is because the iterative EM optimization scheme in BSMAN can gradually correct the initial endmembers.

VII. Conclusion

In this article, we presented a BSMAN for HSIs. An encoder–decoder architecture was designed to incorporate an FCNN with the DIP prior and the forward models to effectively estimate the subpixel labels by learning from both the low-resolution image and prior knowledge without requiring any ground-truth data. The FCNN exploits the spatial correlation effect in the subpixel label field and an efficient purified means approach was adopted to the SPM framework for the endmember estimation. The resulting Bayesian MAP framework is solved by the proposed EM approach. The proposed approach was tested on both real and simulated HSIs, in comparison with several other SPM methods. The proposed BSMAN method was demonstrated effective for the SPM of HSIs with more accurate SPM results than other methods.

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